

Skills Summary

Finding the Missing Factor

Use this reference while completing your worksheet or when studying.

Skill 1 — Missing Factor in an Array

The idea: an array or a set of equal groups shows $(\text{number of rows}) \times (\text{size of one row}) = \text{total}$.

If one of the two factors is missing, **share the total into the part you know:**
 $\text{missing factor} = \text{total} \div \text{known factor}$.

Example: 4 equal rows of counters, 20 counters in all. How many are in each row?

Step 1. The total is shared equally among the rows, so the missing factor is the size of one row — that is a division: total divided by the number of rows.

Step 2. $20 \div 4$: skip-count by 4 — 4, 8, 12, 16, 20 — that is four jumps of 4 to reach 20, so each row holds 5.

Check by multiplying back: $4 \times 5 = 20$ ✓ each row holds 5

Try it: 4 equal rows, 28 counters in all. How many counters are in each row?

check: 7

Skill 2 — Missing Factor from a Fact You Know

Rule: every multiplication fact hides a division fact.

If $a \times b = c$, then $c \div a = b$ and $c \div b = a$.

So to solve $a \times \square = c$, run through the multiples of a until you land on c , and count how many you used.

Example: Solve $8 \times \square = 40$

Step 1. Nothing is shared or drawn here, so use the times table for the factor I know: I want the number of eights that make the total.

Step 2. The eights go 8, 16, 24, 32, 40. The fifth one is 40, so the missing factor is 5.

Related division fact: $40 \div 8 = 5$. $\square = 5$

Try it: Solve $4 \times \square = 28$

check: 7

Skill 3 — Missing Factor in a Story

How to set it up: find the size of one group and the total. Then
(size of one group) $\times$ (number of groups) = total.

The answer's **unit** is whatever the missing factor counts — boxes, trays, rows — not the items inside them.

Example: A bag holds 9 marbles. There are 36 marbles in full bags. How many bags?

Step 1. Each bag is one equal group of the same size, so the number of bags is the missing factor, and I find it by sharing the total into groups of that size.

Step 2. $36 \div 9$: the nines go 9, 18, 27, 36 — four nines make 36.

The missing factor counts bags, so the answer is **4** bags.

Try it: A shelf holds 6 books. There are 30 books on full shelves. How many shelves?

check: 5 shelves

(!) Watch out: the missing factor is never found by multiplying the two numbers you can see. In $8 \times \square = 56$, multiplying gives 448, which is far more than the total — a sure sign the wrong operation was used. Divide, or skip-count up to the total.